D

STM Exercise 9

Phase Diagrams

(Fe-C)

IRON-CARBON (Fe-C) PHASE DIAGRAM

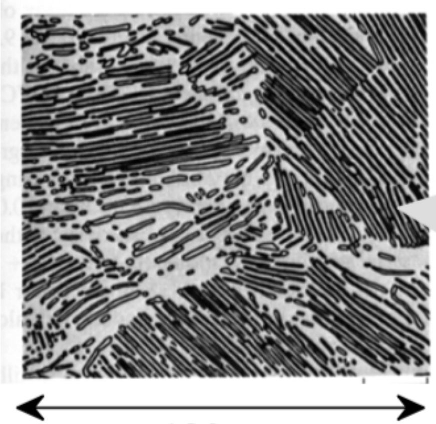
- 2 important points

  -Eutectic (A):

  $L \Rightarrow \gamma + Fe_3C$

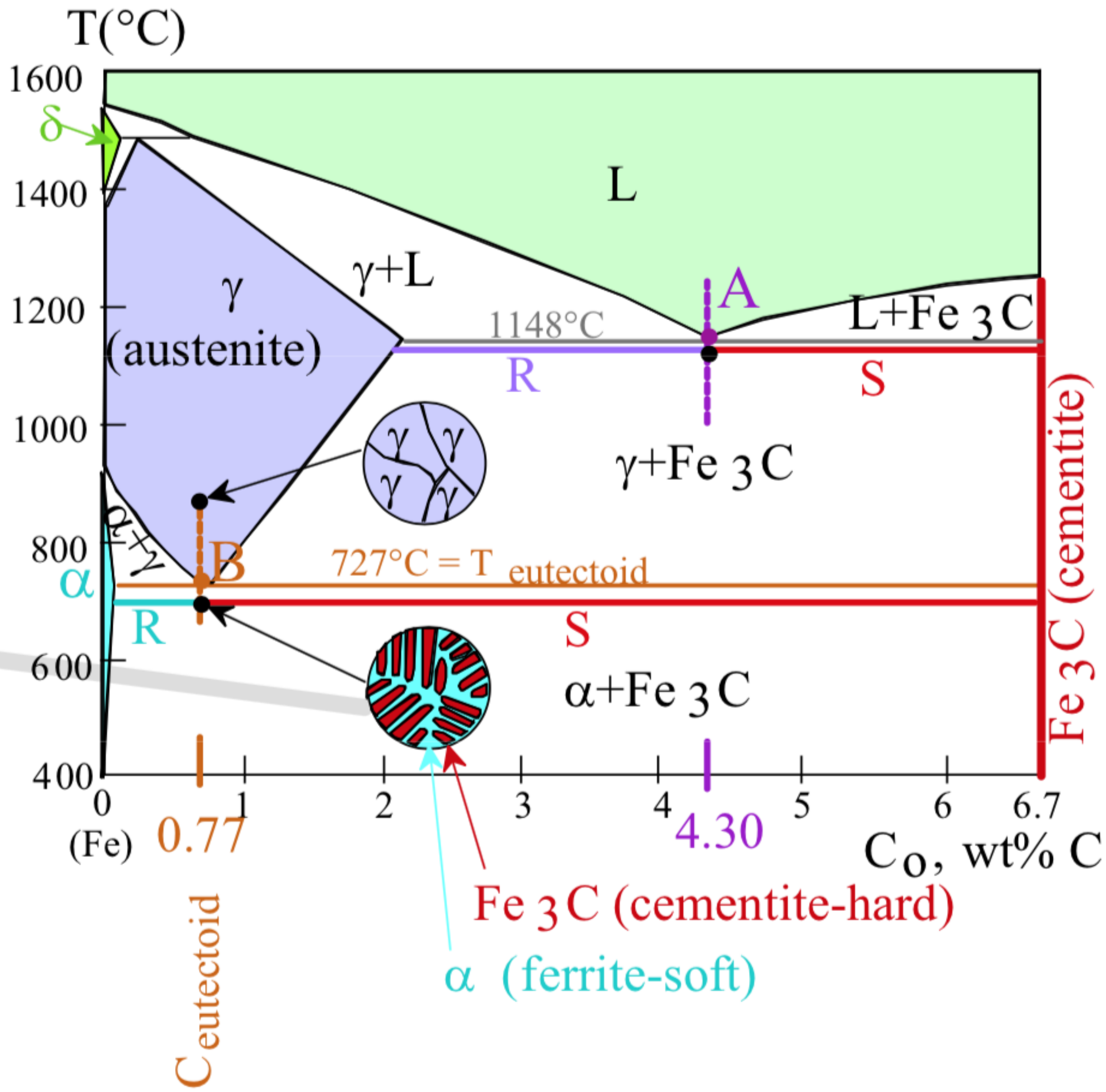
  -Eutectoid (B):

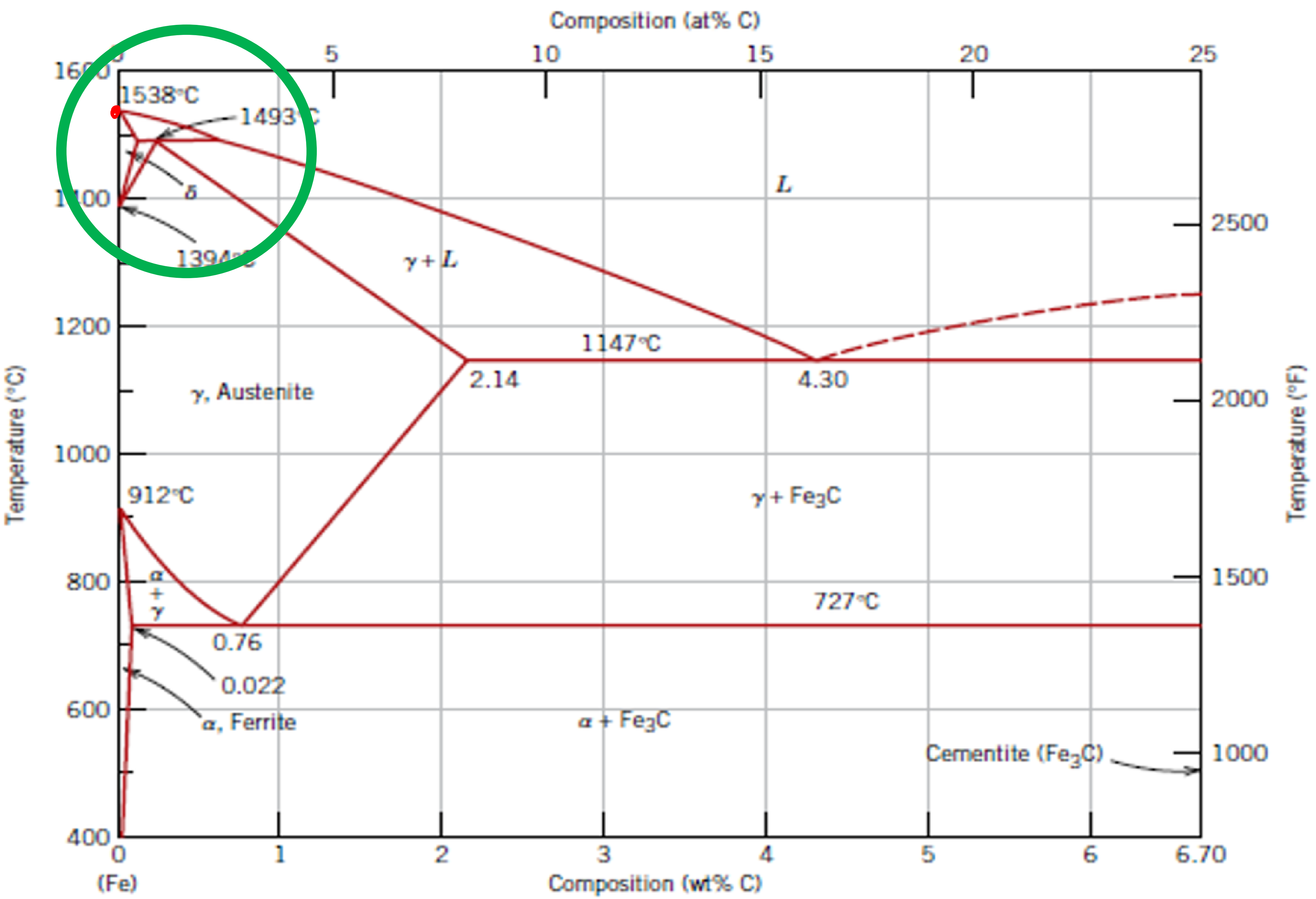
  $\gamma \Rightarrow \alpha + Fe_3C$



120 μm

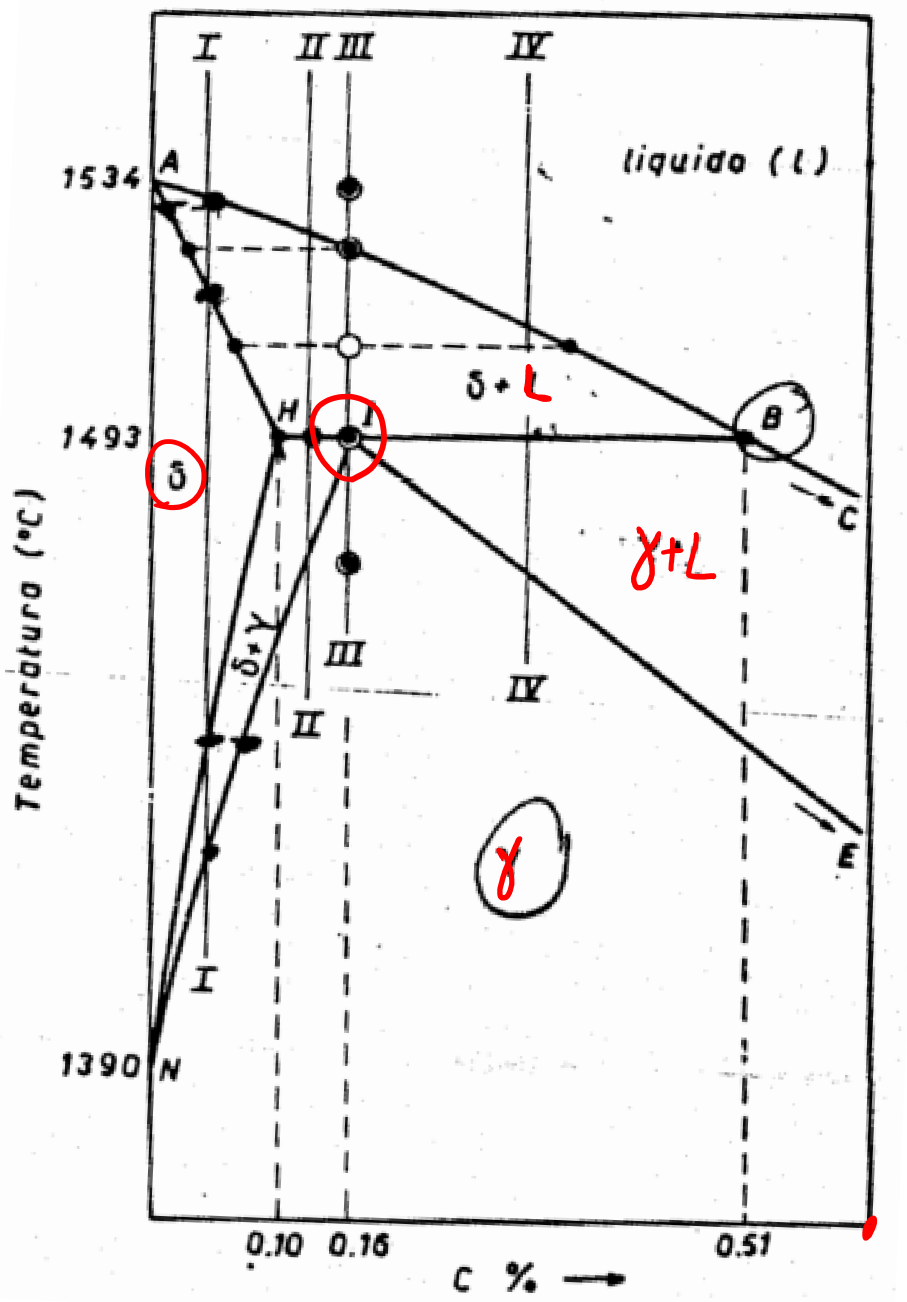
Result: Pearlite = alternating layers of α and Fe_3C phases.





Composition (at% C)
0
5
10
15
20
25
1600
1538°C
1493°C
δ
1400
1394°C
L
γ + L
2500
1200
1147°C
2.14
4.30
γ, Austenite
2000
Temperature (°C)
Temperature (°F)
1000
912°C
γ + Fe3C
800
α
+
γ
727°C
1500
0.76
0.022
600
α, Ferrite
α + Fe3C
Cementite (Fe3C)
1000
400
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)

Exercise 1: which transformation occurs in point I?



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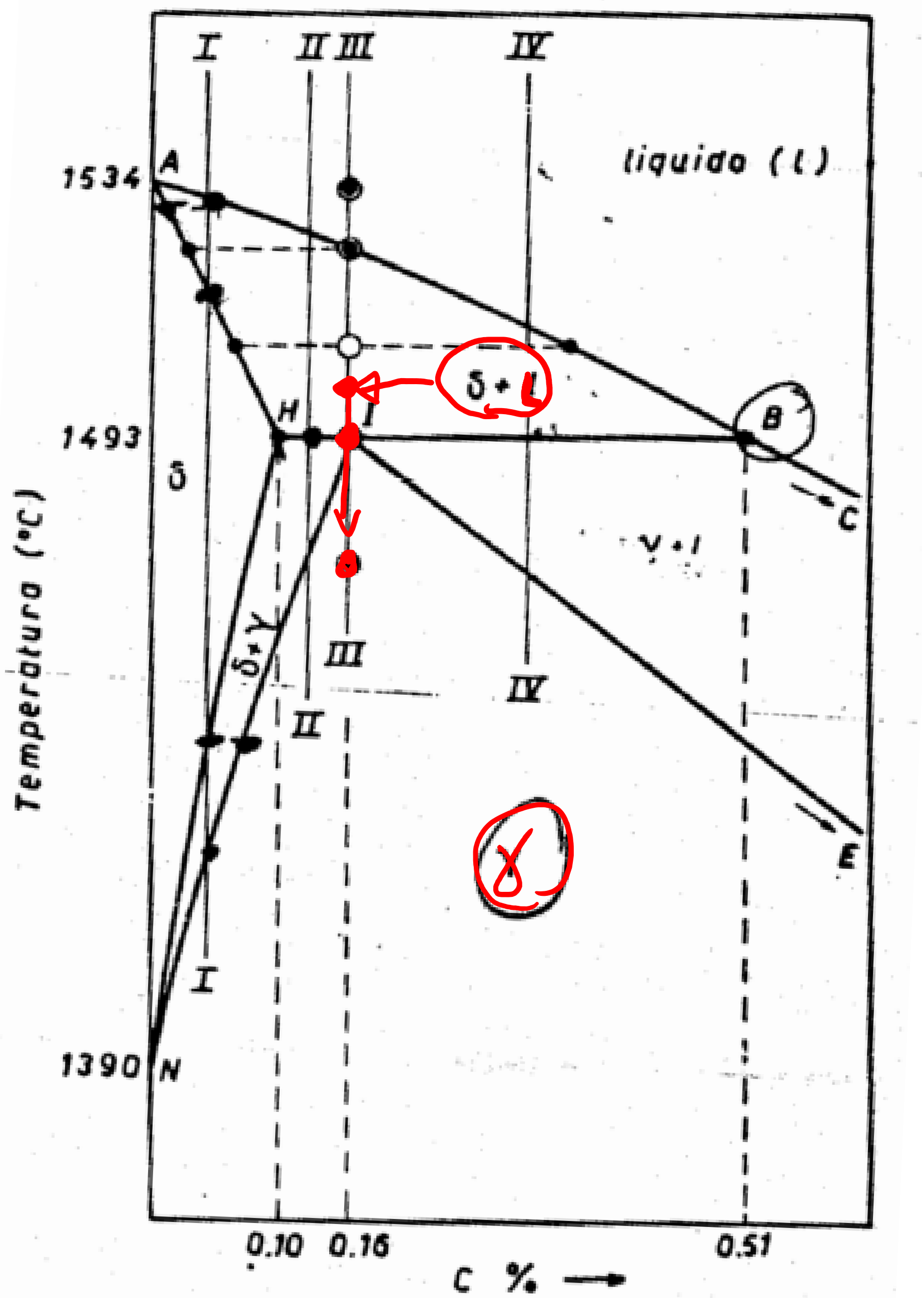

Point I is a **PERITECTIC** transformation

$$\delta + L \leftrightarrow \gamma$$

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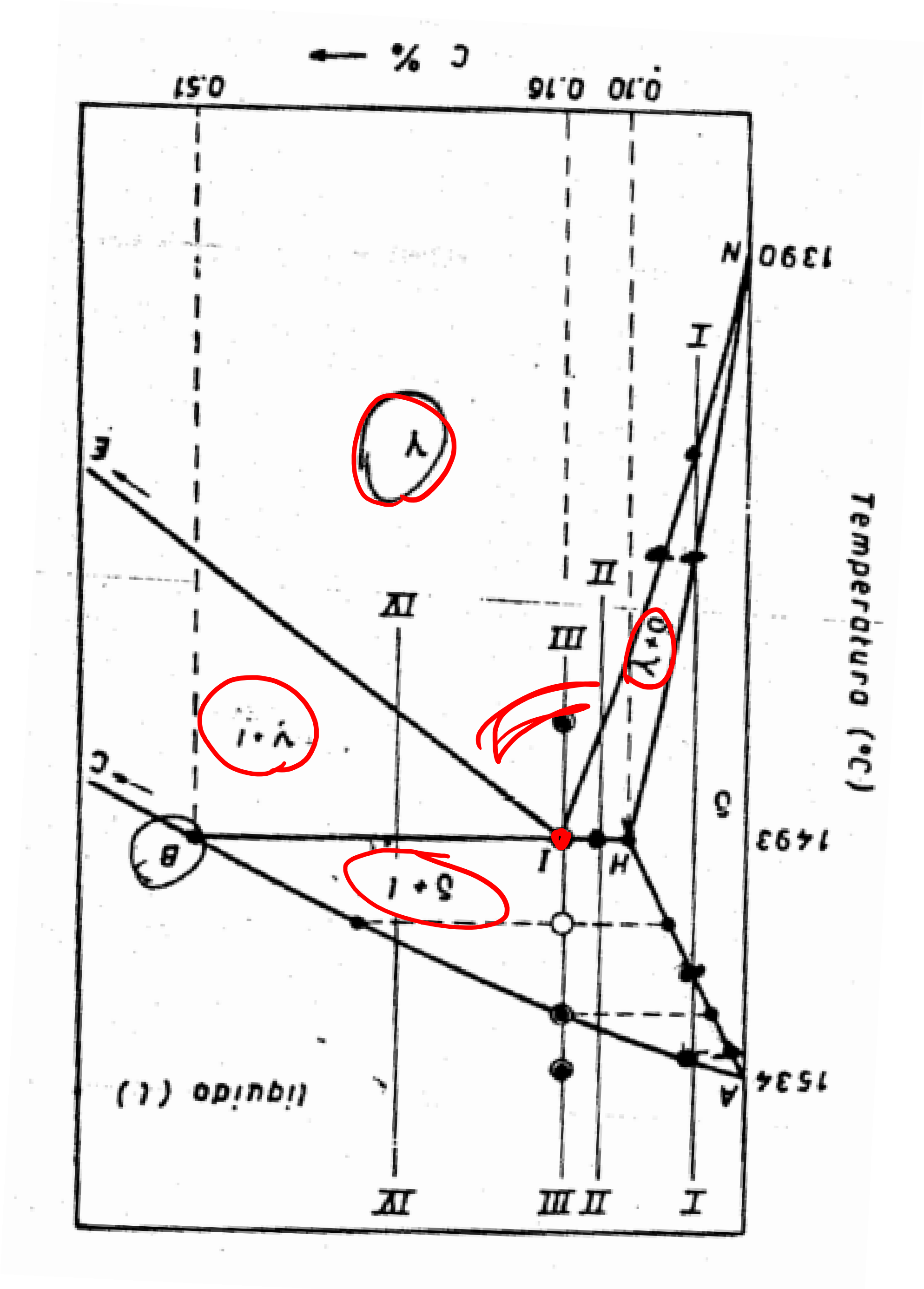
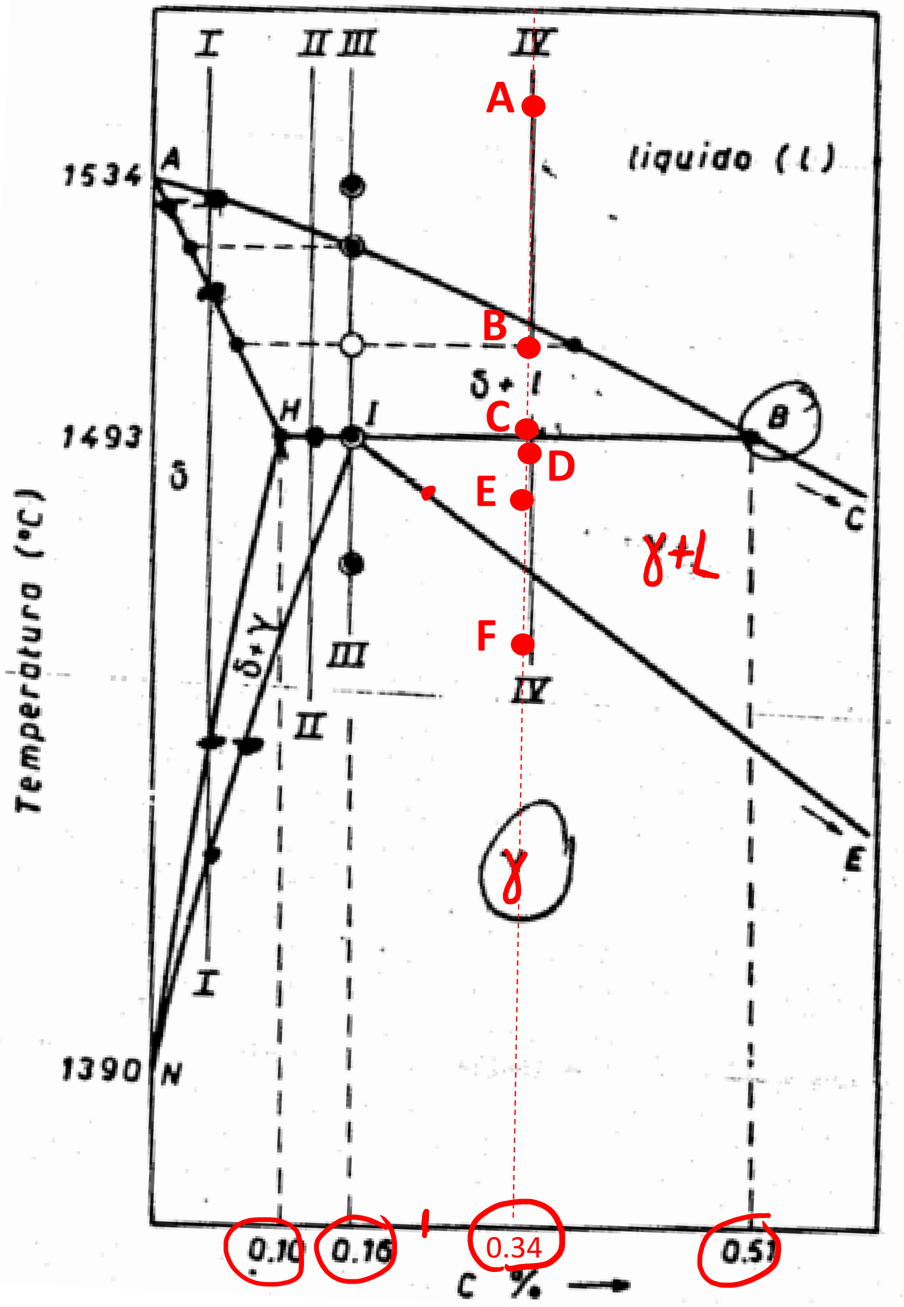
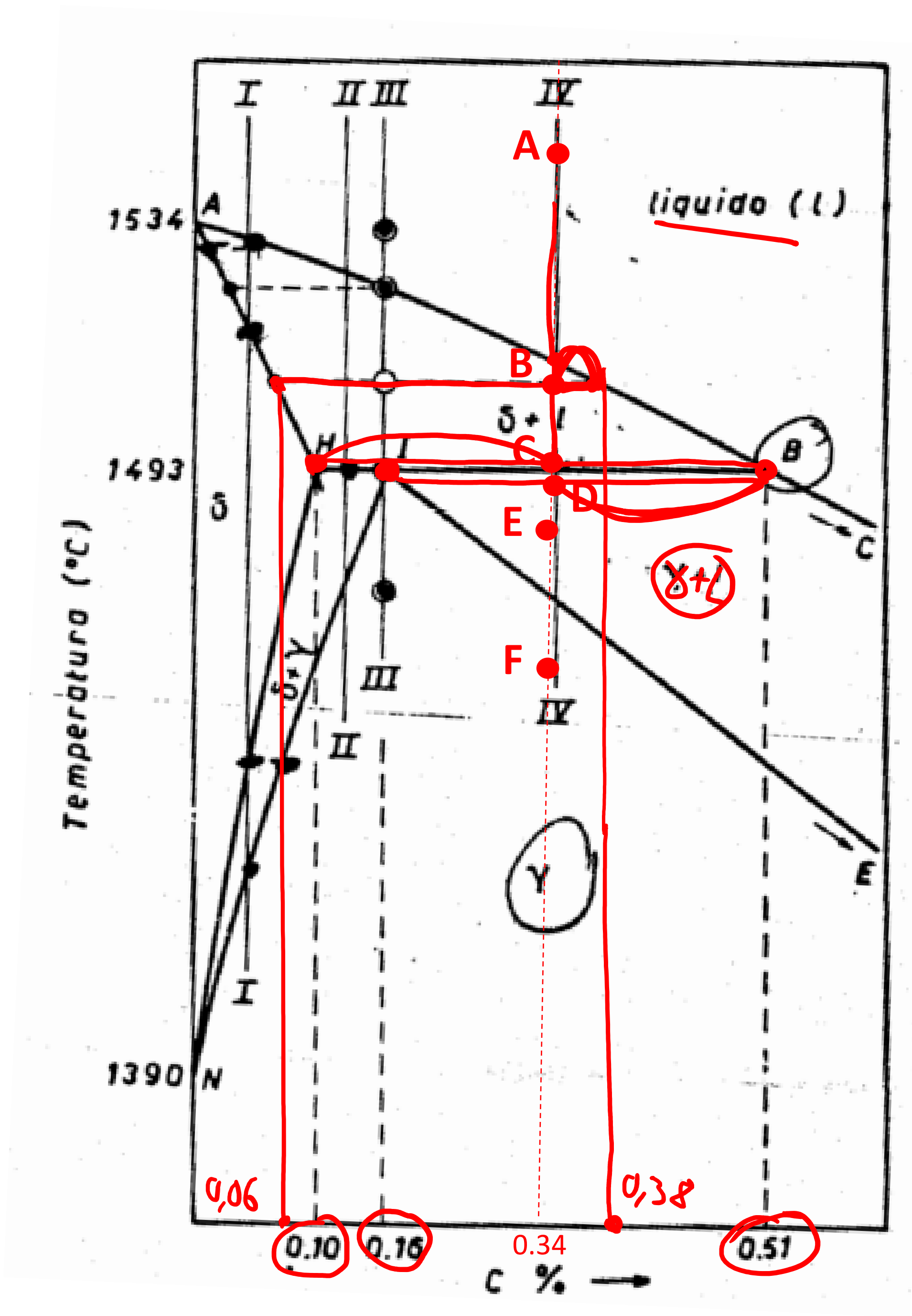


Point I is a **PERITECTIC** transformation

δ+L ↔ γ



Exercise 2: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D (at the end of the peritectic transformation), E and F for a steel with 0.34%C.



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Point (A)	L	
Composition	0,34% C	
Relative amount	100%	

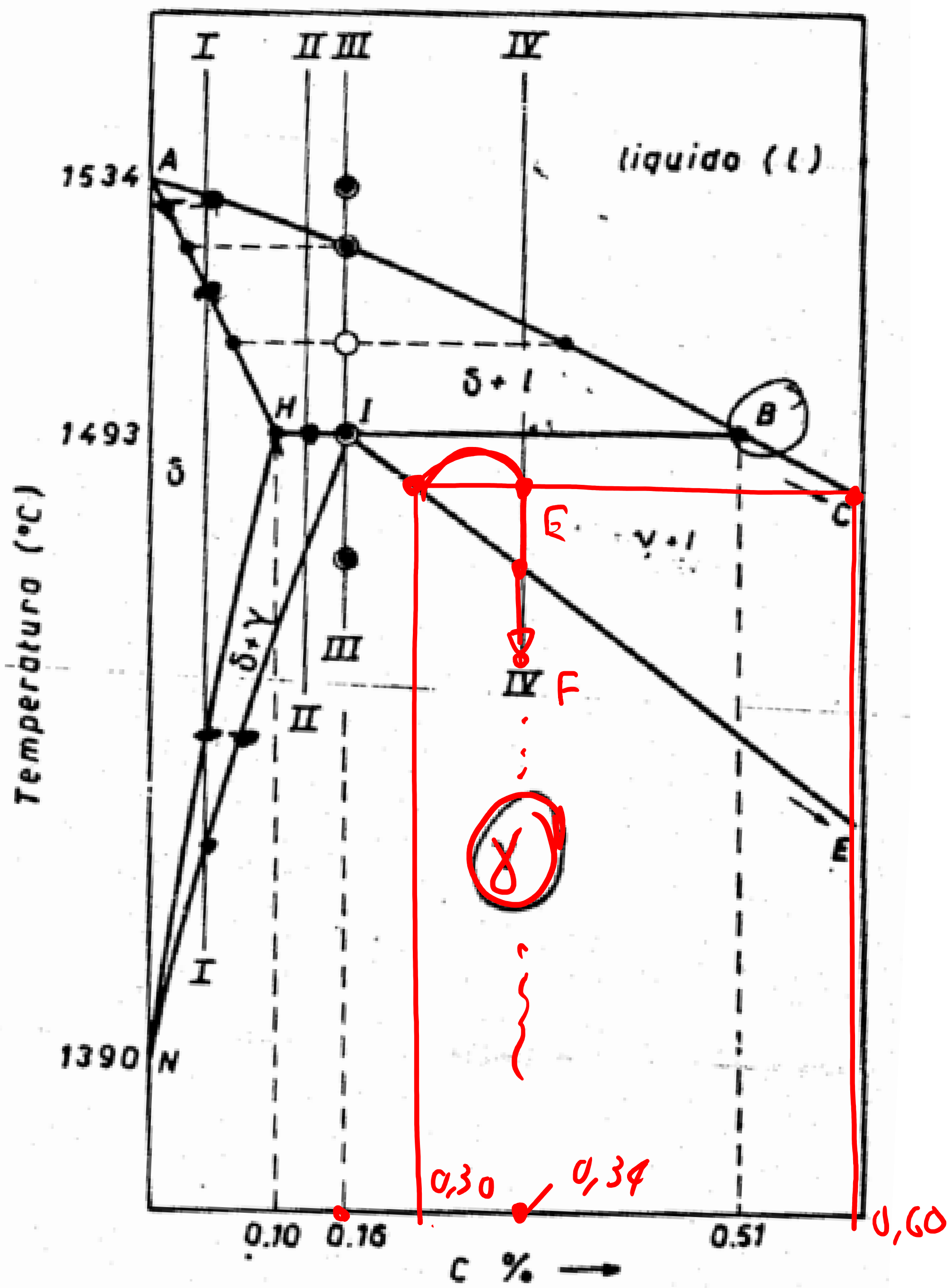
Point (B)	L	δ
Composition	0,38 %C	0,06 %C
Relative amount	87,5%	12,5%

$$\frac{0,38-0,34}{0,38-0,06}\cdot 100$$

Point (C)	L	δ
Composition	0,51%C	0,10 %C
Relative amount	58,5%	41,5%

$$\frac{0,34-0,10}{0,51-0,10}\cdot 100$$

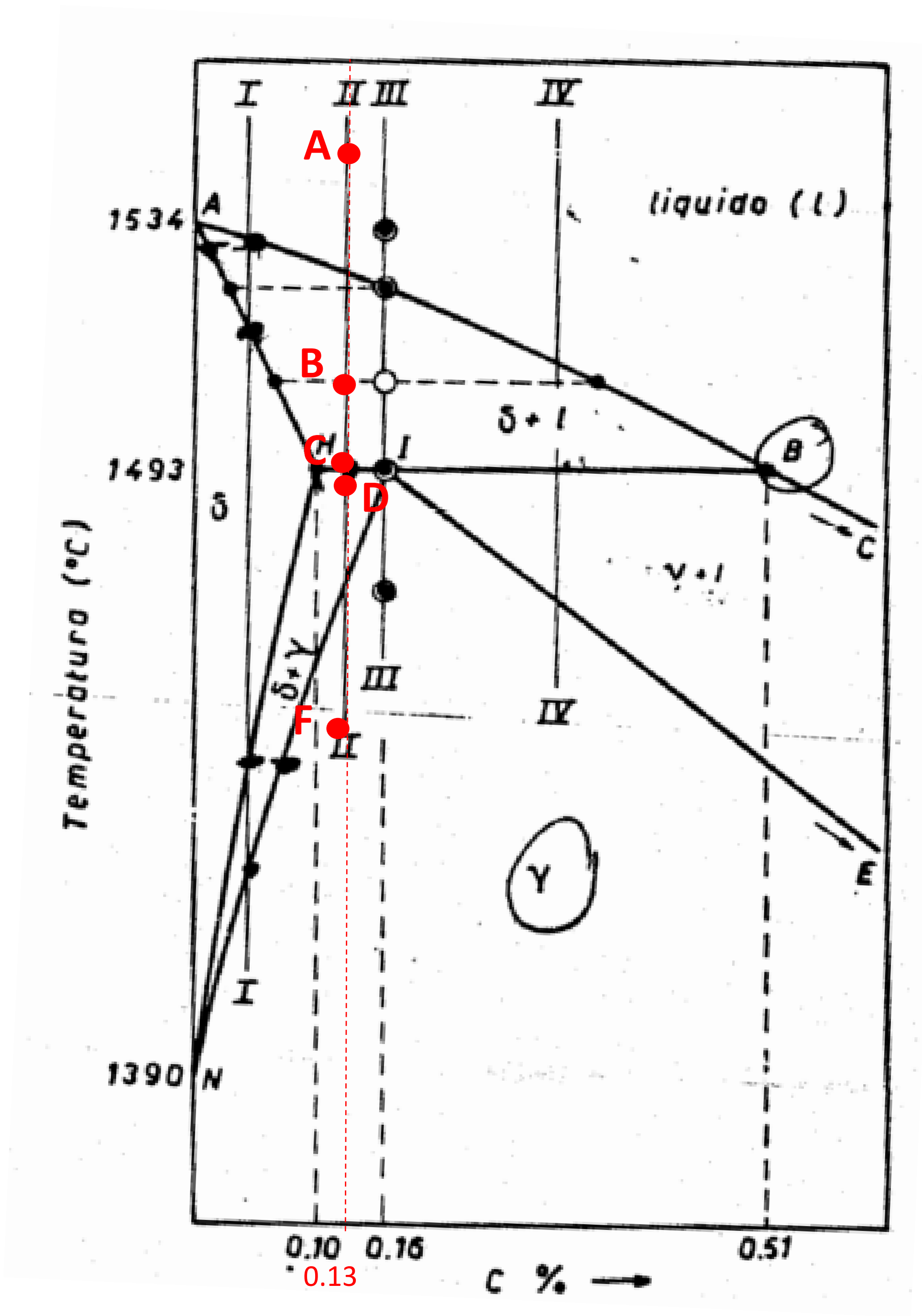
Point (D)	L	γ
Composition	0,51%C	0,16%C
Relative amount	51,4%	48,6%



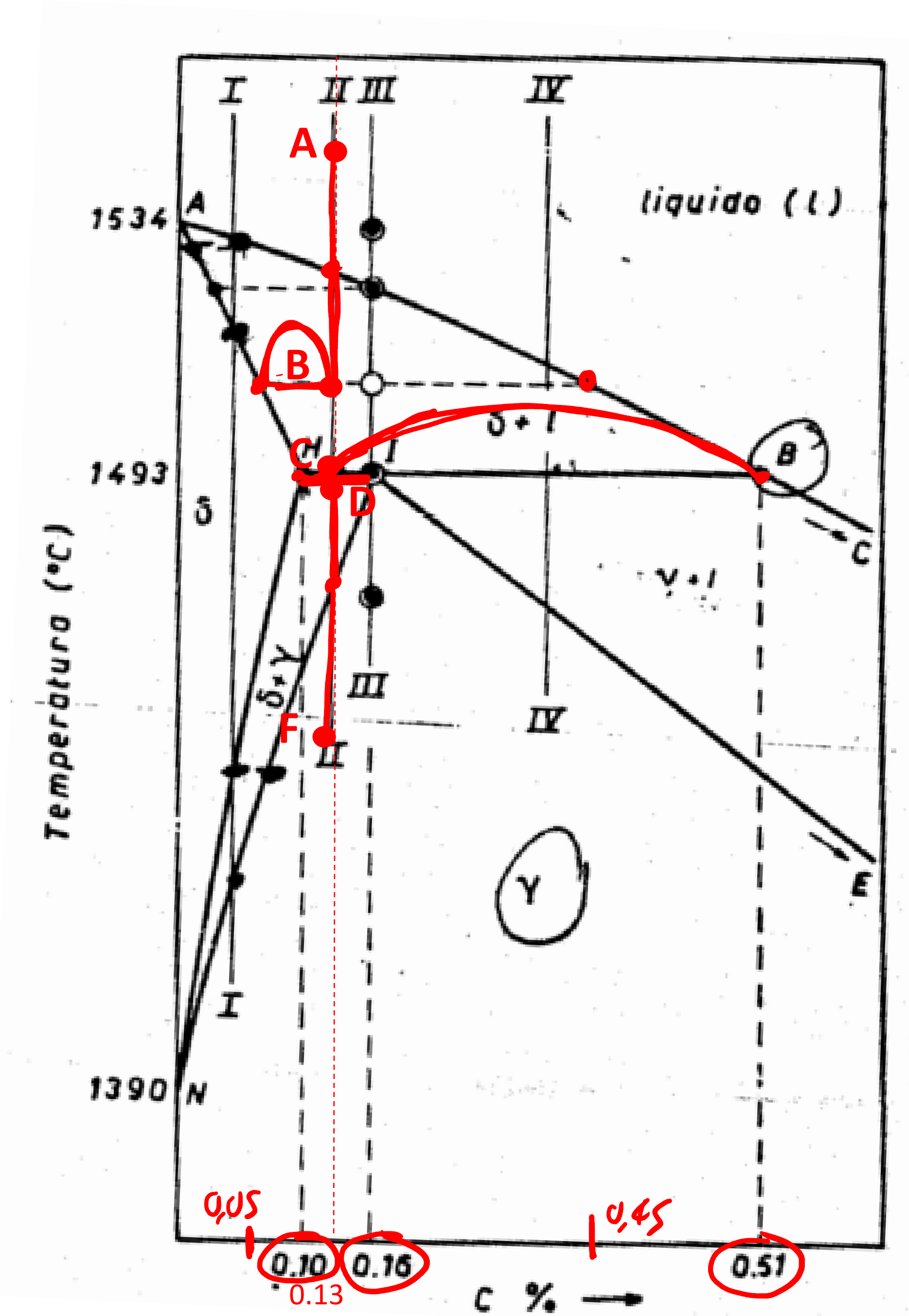
Point (E)	γ	L
Composition	0,30%C	0,60%C
Relative amount	86,7%	13,3%

$$\frac{0{,}34 - 0{,}30}{0{,}60 - 0{,}30} \cdot 100$$

Point (F)	γ	
Composition	0,34%C	
Relative amount	100%	



Exercise 3: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), ~~E~~ and F for a steel with 0.13%C.



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Point (B)	δ	L
Composition	0,05%C	0,45%C
Relative amount	80%	20%

$$\frac{0{,}13-0{,}05}{0{,}45-0{,}05}\cdot 100$$

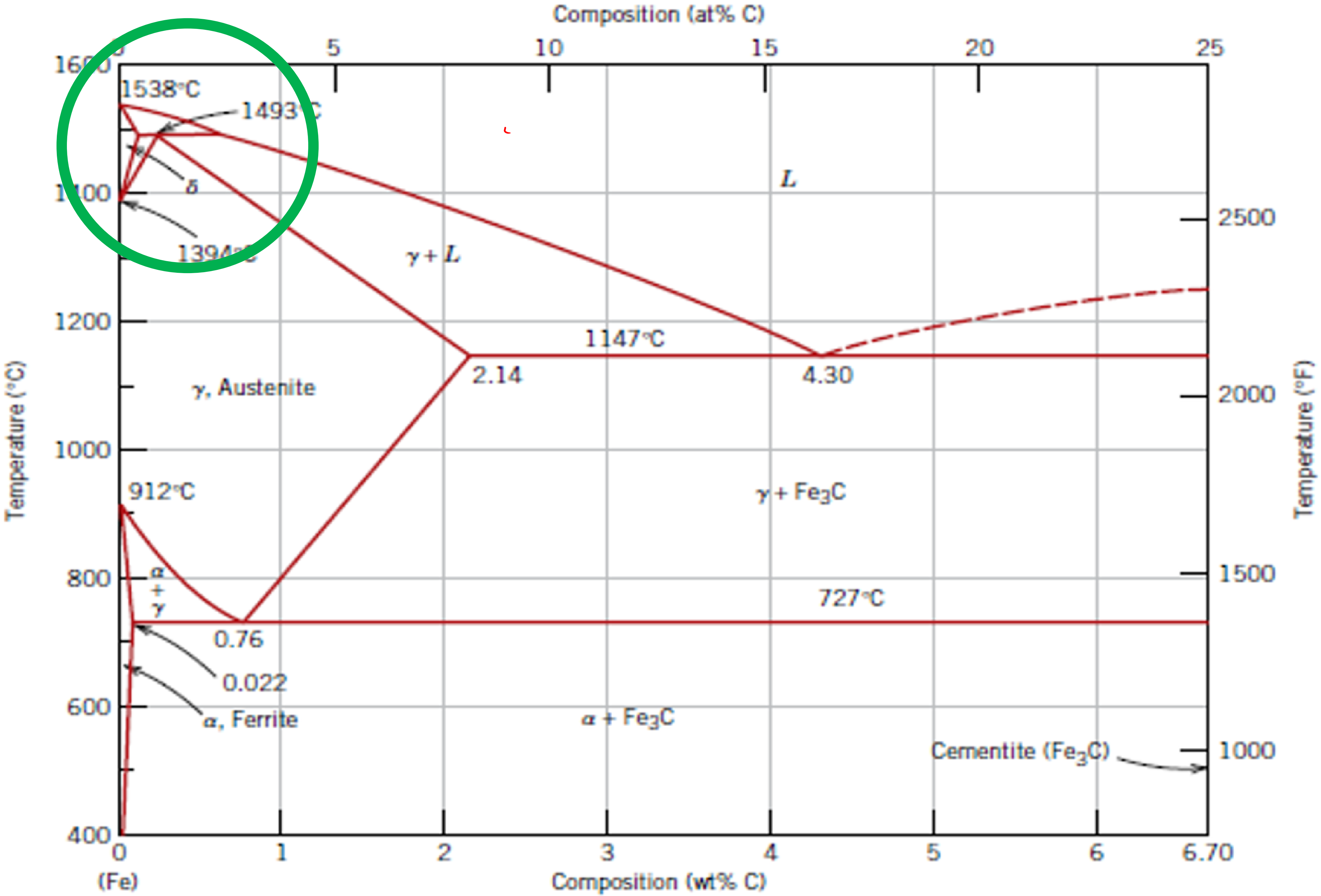
Point (C)	δ	L
Composition	0,10%C	0,51%C
Relative amount	92%	8%

Point (D)	δ	γ
Composition	0,10%C	0,16%C
Relative amount	50%	50%

Point (F)	γ
Composition	0,13%C

END !

Questions ?



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